# HARIRAMAKRRISHNAN RAMACHANDRAN

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## PROFESSIONAL EXPERIENCE — 3.5 YEARS

**Software Engineer — HCL Technologies, Chennai, India** Sep 2021 – Jan 2025

- Developed and maintained Java Spring Boot microservices, designing and implementing **REST APIs** for inter-service communication and data exchange, handling over **3,000 requests per minute**.
- Implemented persistence layers using **Hibernate** and JPA, managing complex entity relationships and optimizing **SQL queries** across PostgreSQL and Oracle databases, reducing average query execution time by **18%**.
- Authored **JUnit** tests and employed **Mockito** for mocking dependencies, achieving over **90% code coverage** for new feature development and ensuring application stability.
- Contributed to the architecture of secure infrastructure systems, collaborating with device engineering teams on the capabilities and efficiency of **HSM-integrated services** for key management.
- Integrated **OpenSSL** libraries into Java applications to perform cryptographic operations, including signing and certificate generation, meeting stringent security requirements for factory provisioning.
- Managed multi-modular Java projects using **Maven**, defining build configurations and dependencies to ensure consistent and reproducible build processes for standalone server applications.
- Containerized Java applications using **Docker** and deployed them to on-premises servers, facilitating consistent environments and simplifying deployment workflows.

## SKILLS

**Languages** – Java, Python, C, SQL, R, XML, JSON

**Frameworks/Libraries** – Spring Boot, Hibernate, JPA, JCA/JCE, RESTEasy, JAXB, JUnit, Mockito, pandas, NumPy, scikit-learn, OpenSSL, PKCS11

**Tools/Platforms** – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, Git, Jira, Confluence, Splunk, Dynatrace, Maven, Gradle, PostgreSQL, MySQL, Oracle, HSM platforms

**Security/Crypto** – PKI, Cryptographic Key Management, Hardware Security Modules (HSMs), Certificate Management, FIPS, JCA/JCE

**Methodologies** – Agile, Scrum, CI/CD, SDLC, Code Reviews, Root Cause Analysis

## PROJECTS

### Secure Key Management Service (HCL Technologies)

- Developed a Java-based server application integrating with **PKCS11-compliant HSMs** to manage cryptographic keys for device image signing operations.
- Implemented **REST APIs** using Spring Boot to provide secure access for internal services requiring key operations, ensuring robust authentication and authorization.
- Containerized the service using **Docker** and configured **Jenkins** pipelines for automated build, test, and deployment to on-premises infrastructure.

### Automated Build and Signing Pipeline (HCL Technologies)

- Engineered a **Python** script to automate the build, signing, and packaging of native applications, integrating with **Maven** build toolchains.
- Integrated **OpenSSL** commands within the pipeline to perform cryptographic signing operations, ensuring integrity and authenticity of the final binary artifacts.
- Managed source code using **Git** and configured the pipeline for automated execution upon code commits, reducing manual intervention by **60%**.

## EDUCATION

**M.Sc. Data Analytics — National College of Ireland, Dublin** Feb 2026

**B.E. Computer Science — SNS College of Technology, Coimbatore, India** 2021

## CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)